One percentage from another

1 Find the original number if:

a 1% of the number is 20.

b 2% of the number is 8.

c 10% of the number is 5.

d 20% of the number is 10.

e 25% of the number is 13.

f 50% of the number is 14.

g 5% of the number is 3.

h 10% of the number is 12.

2 For each of the following:

i Find the 1% of the number.

ii Hence, find the whole number.

a 9% of the number is 72.

b 9% of a number is 63.

c 60% of a number is 480.

d 47% of a number is 188.

e 4% of a number is 236.

f 15% of a number is 7500.

g 12% of a number is 576.

h 8% of a number is 230.

3 For each of the following:

i Find the 1% of the quantity.

ii Hence, find the whole quantity.

a 5% of a quantity is $\$320$.

b 11% of a quantity is $\$16.50$.

c 30% of a quantity is 2700 mL.

d 54% of a quantity is 6210 mL.

e 14% of a quantity is 7280 g.

f 22% of a quantity is $88\,000$ g.

g 34% of a quantity is 918 L.

h 78% of a quantity is $10\,959$ cm.

4 For each of the following:

i Find 10% of the quantity.

ii Hence, find the whole quantity.

a 150% of a quantity is 135.

b 240% of a quantity is 1464.

c 80% of a quantity is $\$160$.

d 320% of a quantity is $\$13\,280$.

5 For each of the following:

i Find the 25% of the quantity.

ii Hence, find the whole quantity.

a 125% of a quantity is 30.

b 50% of a quantity is 85.

c 150% of a quantity is $75\,000$.

d 250% of a quantity is 1500.

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6 For each of the following, find the whole quantity:

a 14% of a quantity is 637.

b 26% of a quantity is $10\,400$.

c 32% of a quantity is $\$10\,240$.

d 120% of a quantity is $\$87.30$.

e 80% of a quantity is $840\ \text{mL}$.

f 57% of a quantity is $4651.2\ \text{g}$.

g 64% of a quantity is $\$10\,400$.

h 92% of a quantity is $4738\ \text{m}$.

7 If 36% of a quantity is equal to 2412:

a Find 1% of the quantity.

b Find 83% of the quantity.

8 If 6% of a number is equal to 36:

a Find 1% of the number.

b Find 36% of the number.

9 If 21% of a quantity is equal to 378:

a Find 1% of the quantity.

b Find 71% of the quantity.

10 870% of a number is 696. Find the original number.

11 If 48% of 85 is 40.8, find 85% of 48.

Applications

12 $33\frac{1}{3}\%$ of a quantity is 448 grams. Find $66\frac{2}{3}\%$ of the same quantity.

13 462 students represent 30% of the school population. Find the number of students that make up 60% of the school population.

14 A city has a yearly water supply of $27\,000\,000$ megalitres. It gets 14% of this water supply from melted snow. Find the amount of water that comes from melted snow each year.

15 When a rock sample was examined, it was found that 4.1 kilograms of it was copper. If this sample represents 10% of the total copper in the rock bed, how many kilograms of copper were there in the rock bed?

16 The 64 girls at Victoria's party represent 80% of all the guests. Find the total number of guests at the party.

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17 Beth spends 90% of her weekly income on rent. If her rent is $\$720$ each week, find her weekly income.

18 After 10% GST was added, the price of an iPod was $\$238$. Calculate the original price before GST.

19 A total of $36\,000$ engineers were hired last year. This represents 9% of the total number of engineers employed in the country. How many engineers are employed in the country altogether?

20 Iain spent some time volunteering in a community. He spent 10% of his time volunteering at schools and 2% of his time volunteering at the local hospital. The remaining 132 days he spent volunteering at animal shelters.

How many days did Iain volunteer at the local hospital?

21 Oprah has completed 50% of the necessary 60 hours of pilot training.

a Find 50% of 60 hours.

b Ryan has completed 5% more of the required training hours than Oprah. How many hours has he completed?

22 Fred's new cable television bill is $\$98$ per month. This is 140% of his monthly bill from last year, when he had the basic cable package. What was his monthly cable television bill last year?

23 Gwen has $\$200$ to spend on a new laptop. The sale price of the laptop includes the original price plus a sales tax of 9%. Find the greatest original price of the laptop that Gwen can afford. Round your answer to the nearest dollar.

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